MOHAMED EID SHEHATA

Data Scientist | AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer (2026 graduate) with hands-on experience in machine learning, deep learning, NLP, computer vision, and LLMs. Trained deep learning models reaching 98.6% R^2 and under 2% MAPE, and built NLP and object detection systems benchmarked against strong baselines. Experienced in taking projects from data preparation to deployment with Streamlit, Gradio, and ONNX, and explaining results with SHAP and Power BI.

TECHNICAL SKILLS

- **Machine Learning & Statistics:** Classification, Regression, Clustering, Predictive Modeling, Data Preprocessing, Feature Engineering, Model Evaluation & Optimization, Statistical Analysis, A/B Testing, XAI (SHAP)
- **Generative & Agentic AI:** LLMs, RAG (Corrective RAG), AI Agents, Multi-Step Reasoning, Tool Integration (Web Search), Workflow Orchestration (LangGraph), Prompt Engineering, Hallucination Checking
- **Deep Learning, NLP & Computer Vision:** CNNs, RNNs (LSTMs, GRUs, BiLSTM), Transfer Learning (DistilBERT), NLP Pipelines, Object Detection (YOLO), Monte Carlo Dropout
- **Programming, Databases & APIs:** Python (Pandas, NumPy), SQL (MySQL, SQL Server), ChromaDB (Vector DB), FastAPI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenCV, YOLO
- **Evaluation, MLOps & Deployment:** RAGAS, LangSmith, Docker, FastAPI, Streamlit, Gradio, ONNX, Git, GitHub, Azure
- **Data Analytics & BI:** Power BI, Tableau, ETL & Data Pipelines, EDA, Data Visualization, Actionable Insights
- **Soft Skills:** Analytical Thinking | Cross-Functional Collaboration | Stakeholder Communication | Problem-Solving
- **Languages:** Arabic (**Native**) | English (**Intermediate**)

PROFESSIONAL EXPERIENCE

Instant Software Solutions | Artificial Intelligence Trainee **Jul 2026 – Aug 2026**

- **Completed** an intensive AI & Data Science program with projects in AI agents, NLP, Computer Vision, GANs, and data analysis.

NTI Creativa | Data Analysis Trainee **Mar 2026 – May 2026**

- **Collected** structured data from multiple sources into Excel, SQL Server, and Power BI with consistent, error-free records.
- **Cleaned** raw data by fixing inconsistencies, duplicates, and missing values before analysis.

PROJECTS

Deep Learning Surrogate Models for Solid Rocket Motor Grain Design (Graduation Project) **Jul 2026**

Tools: Python, TensorFlow, Keras, CNN, LSTM/GRU, Monte Carlo Dropout, Streamlit

- **Engineered** 18 deep learning pipelines (custom CNN, cross-attention) to replace CFD simulations with millisecond-level inference, achieving 98.6% R^2 and sub-2% MAPE while integrating Monte Carlo Dropout for uncertainty quantification.

Customer Complaint Classification (Deep Learning & NLP) **Jul 2026**

Tools: Python, DistilBERT, GRU/LSTM, Gradio | GitHub Repo

- **Engineered** an NLP pipeline for 162K+ imbalanced records, benchmarking DistilBERT (85.2% F1) against a resource-optimized GRU (82.9% F1) which was deployed via Gradio for cost-efficient, high-speed real-time inference.

Customer Churn & Profit Optimization (ML + XAI + BI) **Mar 2026**

Tools: Python, Scikit-learn, SHAP, Streamlit, Power BI | GitHub Repo

- **Developed** a revenue-driven ML pipeline addressing $3.68M churn exposure via a custom Profit-Based Priority Score, deploying a Streamlit app with real-time SHAP (XAI) explanations and a companion Power BI dashboard.

Road Damage Detection (Object Detection & Benchmarking) **Aug 2026**

Tools: Python, YOLO (Ultralytics), ONNX, Streamlit, OpenCV | GitHub Repo

- **Developed** a real-time defect detection system by benchmarking YOLO architectures, exporting the optimal YOLOv8n (89.8% mAP50, ~104 FPS) to ONNX, and deploying it as an interactive Streamlit app for automated video processing.

EDUCATION

Bachelor's Degree in Navigation Science & Space Technology

Beni Suef University, Egypt **Graduated 2026**

CERTIFICATIONS

- **Data Science & AI: AI & Data Science Diploma** (Instant Software Solutions, **Aug 2026**) | **Associate Data Scientist** (DataCamp, **Aug 2025**) | **Data Science & Machine Learning** (Edraak, **Oct 2024**)
- **Machine Learning & Cloud:** Machine Learning (Coursera, **Nov 2025**) | **AI Foundations Associate** (Oracle, **May 2025**)
- **Data Analytics & SQL: Google Data Analytics** (Coursera, **Sep 2024**) | **Data Analyst Track** (DataCamp, **May 2025**)
  SQL & Advanced SQL (365 Data Science, **Nov 2024**)